Embedded Controller

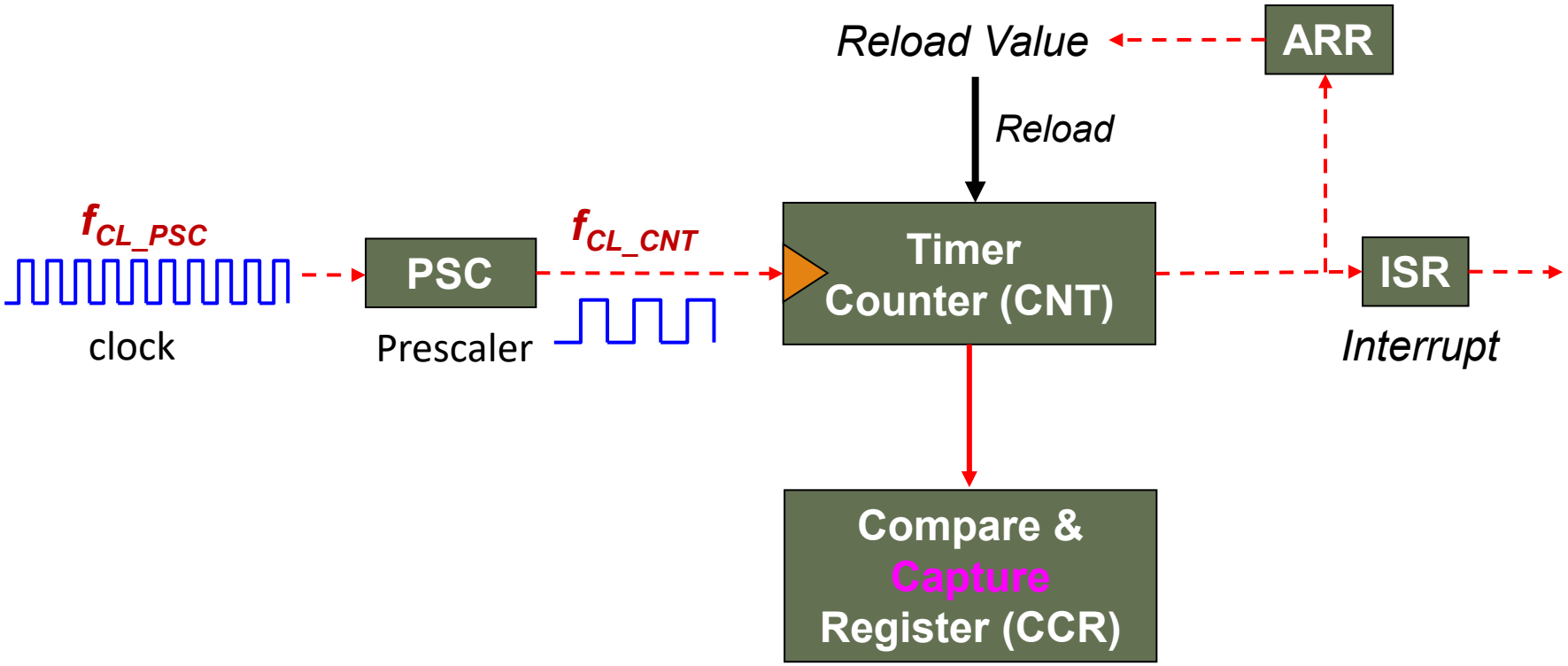
Timer : Input Capture

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Handong Global University

Input Capture

- **Previous Lecture**
  - Basics of Timer
  - Applications of Timer
    - **Timer Interrupt:** **No Output**
    - **PWM:** **Output**
    - **Input Capture:** **Input**

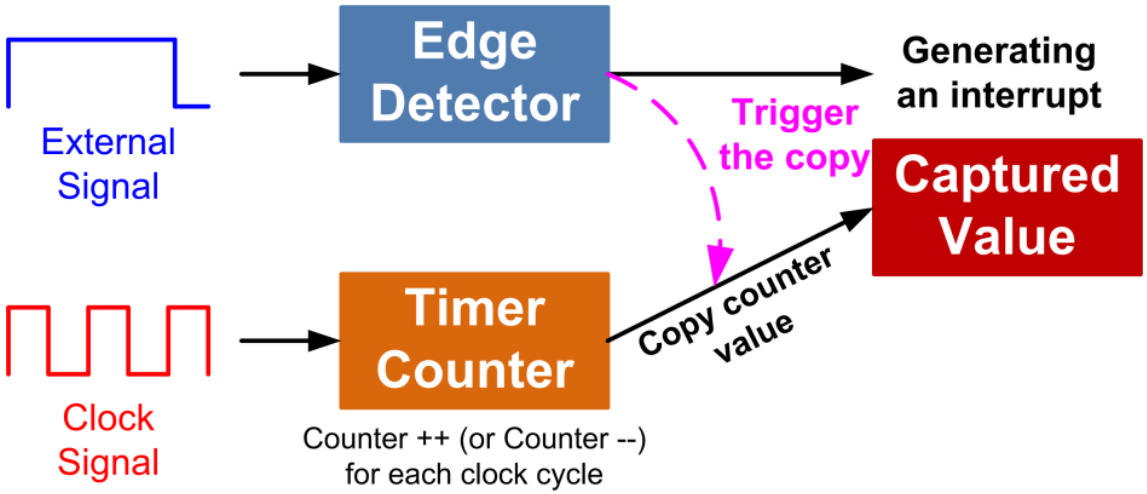


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Input Capture

- **Input Capture**
  - At edge of trigger signal, captures current counter vale
    - When triggered, counter(CNT) value is copied to CCR
    - Difference between two CCR values → measure time span

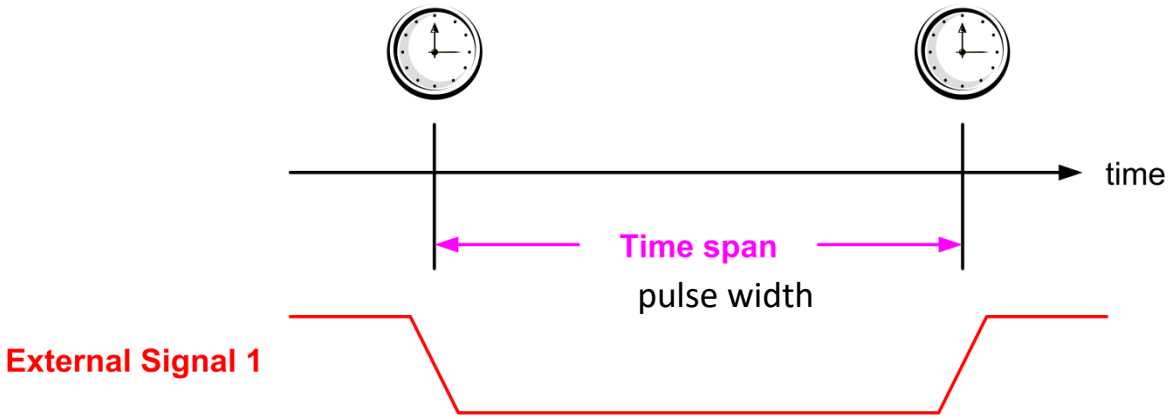
  - Application:
    - Time capture
    - Ultrasonic pulse capture
    - Encoder



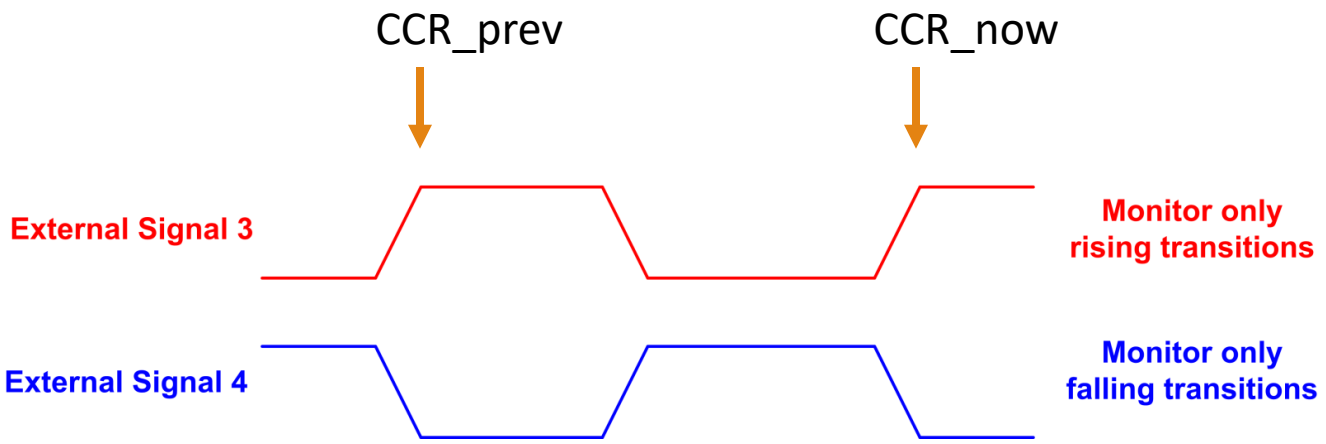
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Input Capture

- **Input Capture**
  - Capture mode 1: Both Rising and Falling



- Capture mode 2: Rising or Falling Edge Only



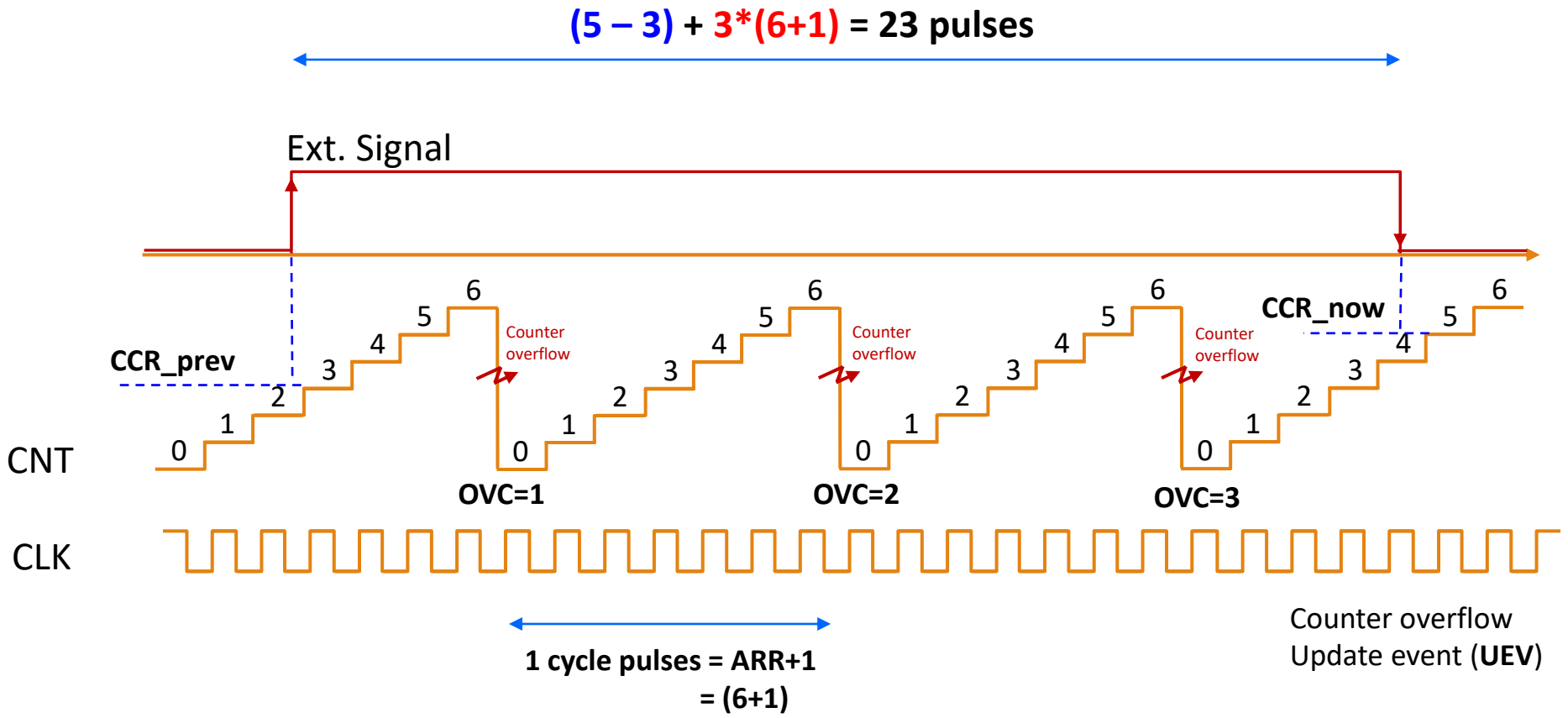
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Input Capture

- **Time Span:**

time span= CNT_CLK_period * [(CCR_{now}-CCR_{prev}) + OVC*(ARR+1)]

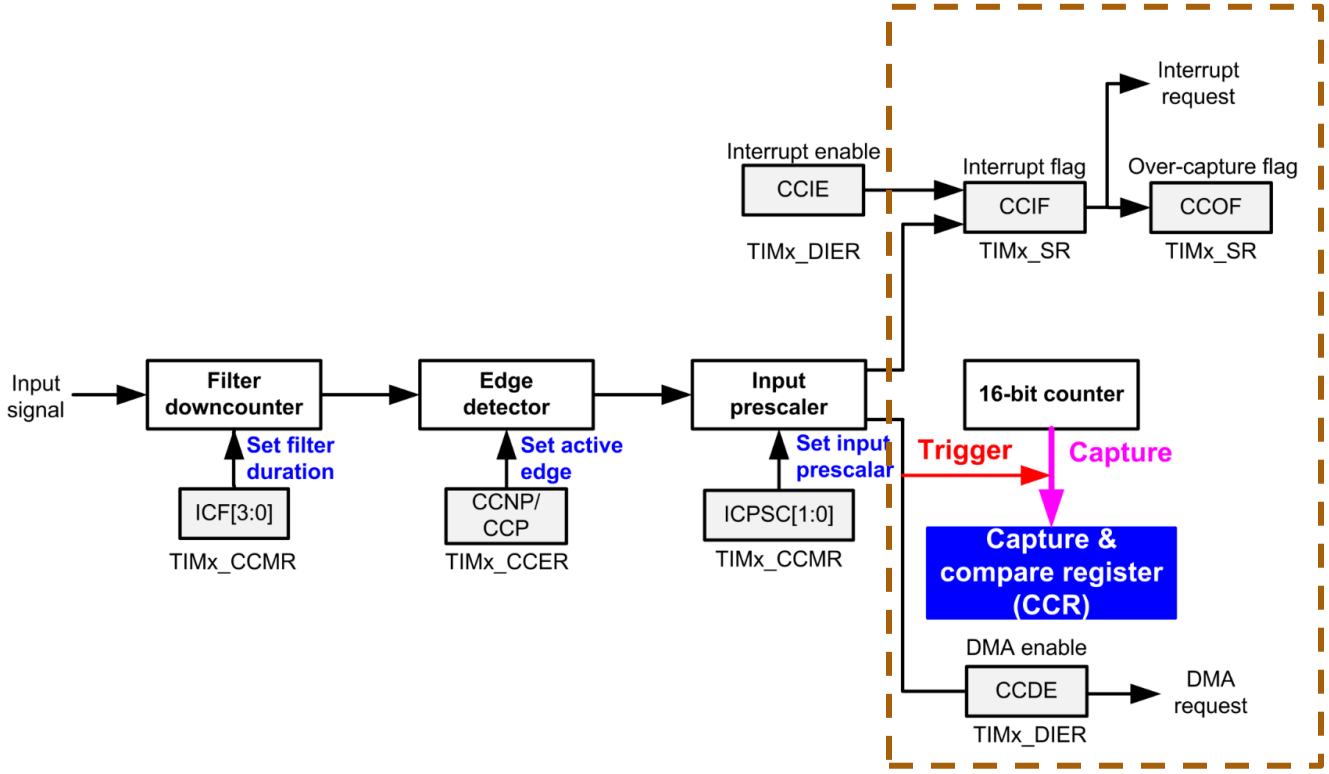
OVC: Overflow Counter



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Input Capture

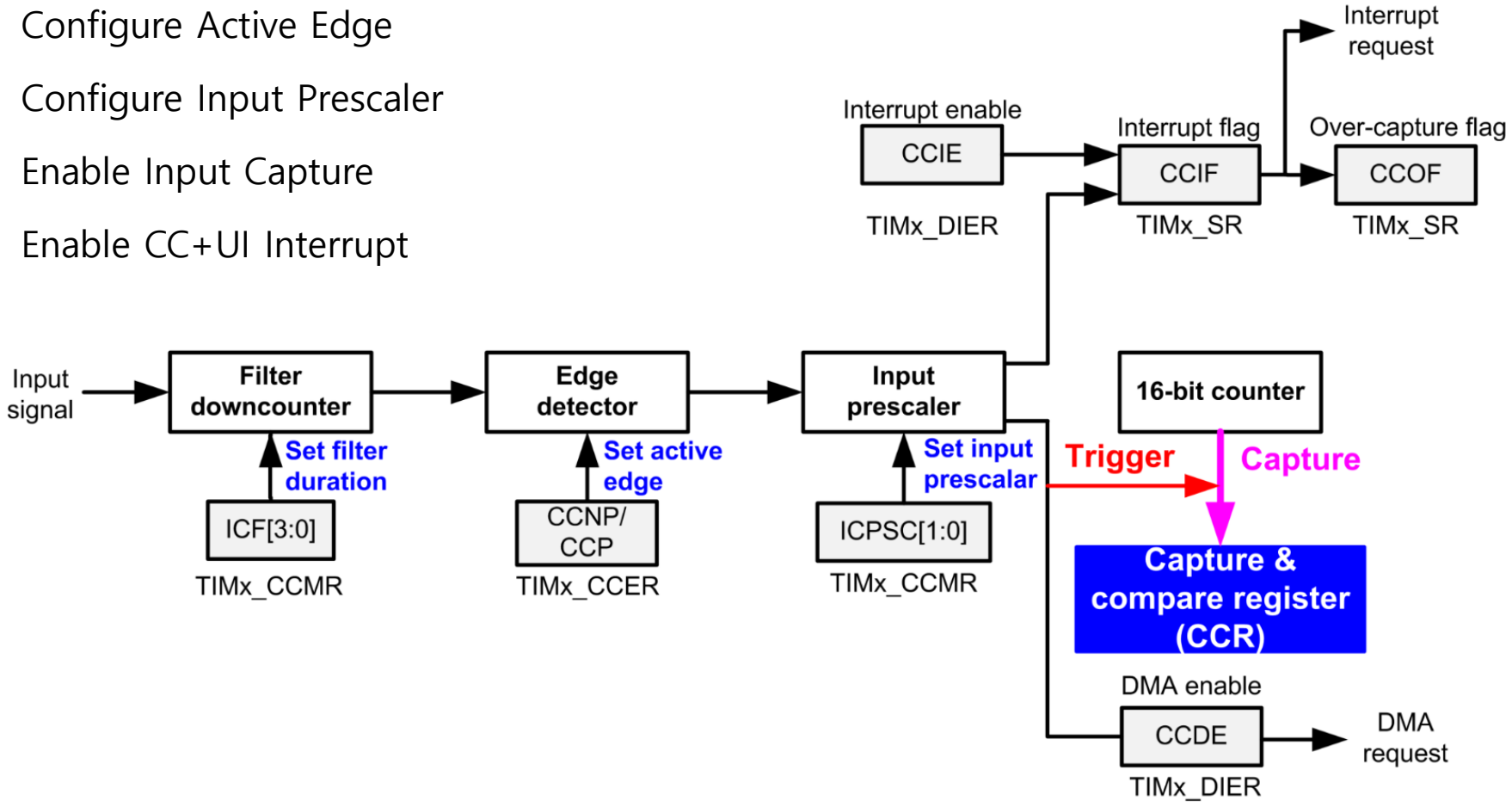
- **Input Capture Occurs**
  - When external signal edge is detected
    - Instantly **CCR** captures **CNT** value
    - Capture Interrupt Flag is set **CCIF =1** with enabled **CCIE=1**
      - Flag is cleared **CCIF=0** by reading **CCR**
    - Overflow Interrupt Flag is set **CCOF=1,** if overflow event occurs (while CCIF=1)
    - Manually record OVC numbers



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Input Capture

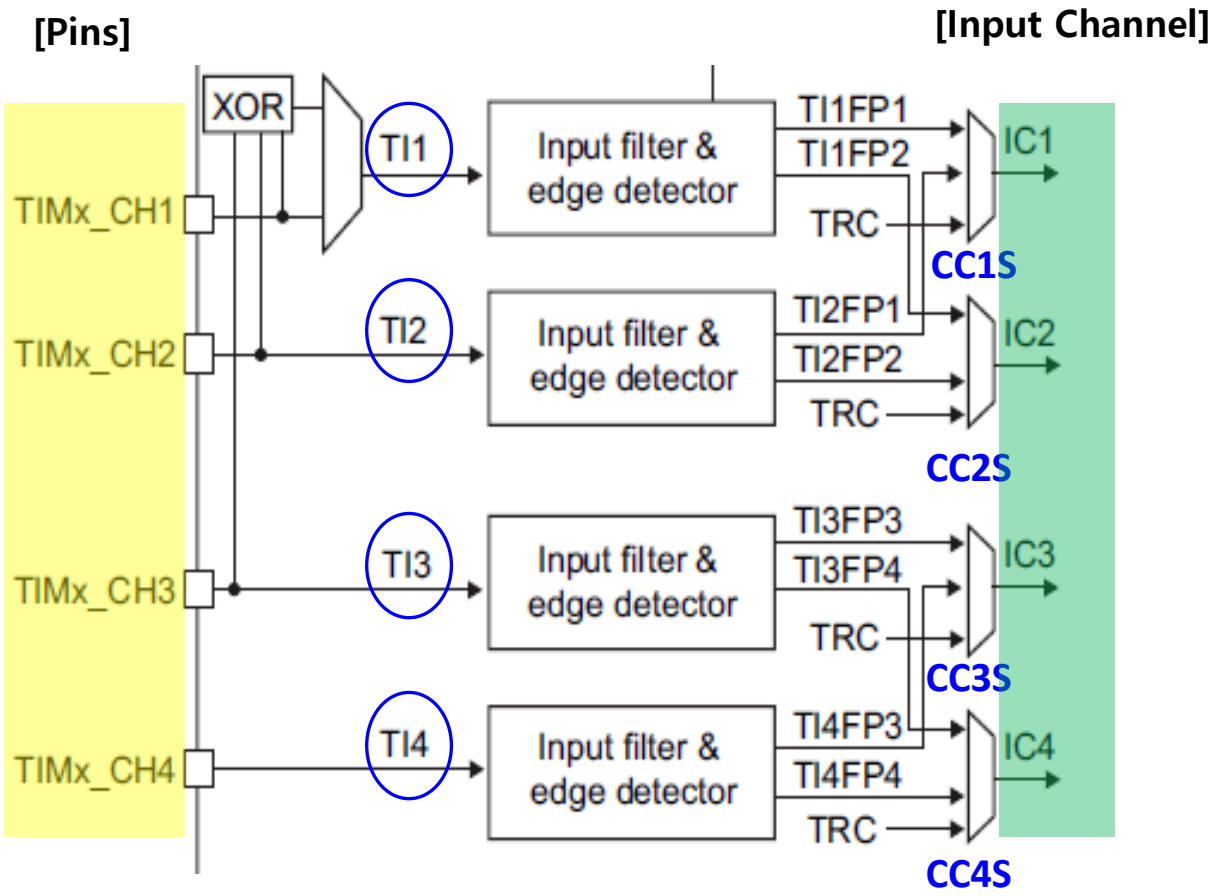
- **Configuration of Input Capture**
  1. Select Input Signal
  2. Configure Input Filter
  3. Configure Active Edge
  4. Configure Input Prescaler
  5. Enable Input Capture
  6. Enable CC+UI Interrupt



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Input Capture: Choosing Input

- **Step 1: Choose Time Input Signal**
  - Each Timer (TIMx) has up to
    - 4* Timer Inputs: TI1 ~ TI4
    - 4* Input Capture Channel: IC1 ~ IC4



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Input Capture: Choosing Input

- **Step 1: Choose Time Input Signal**
  - Each Timer (TIMx) has up to 4 channels, 4 TimerInput (TI), 4 InputCapture (IC)

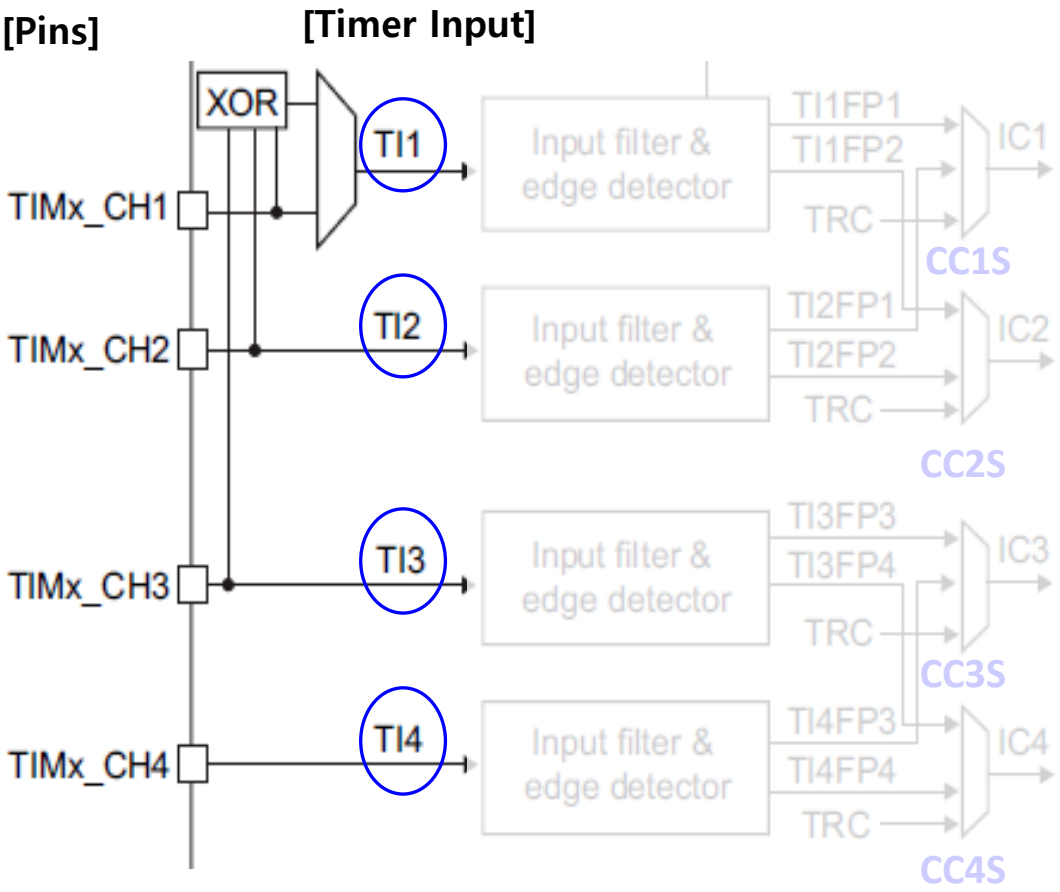
[Pin - Timer Input]

TI1 can be from **TIx_CH1~3**

TI2 is from **TIx_CH2**

TI3 is from **TIx_CH3**

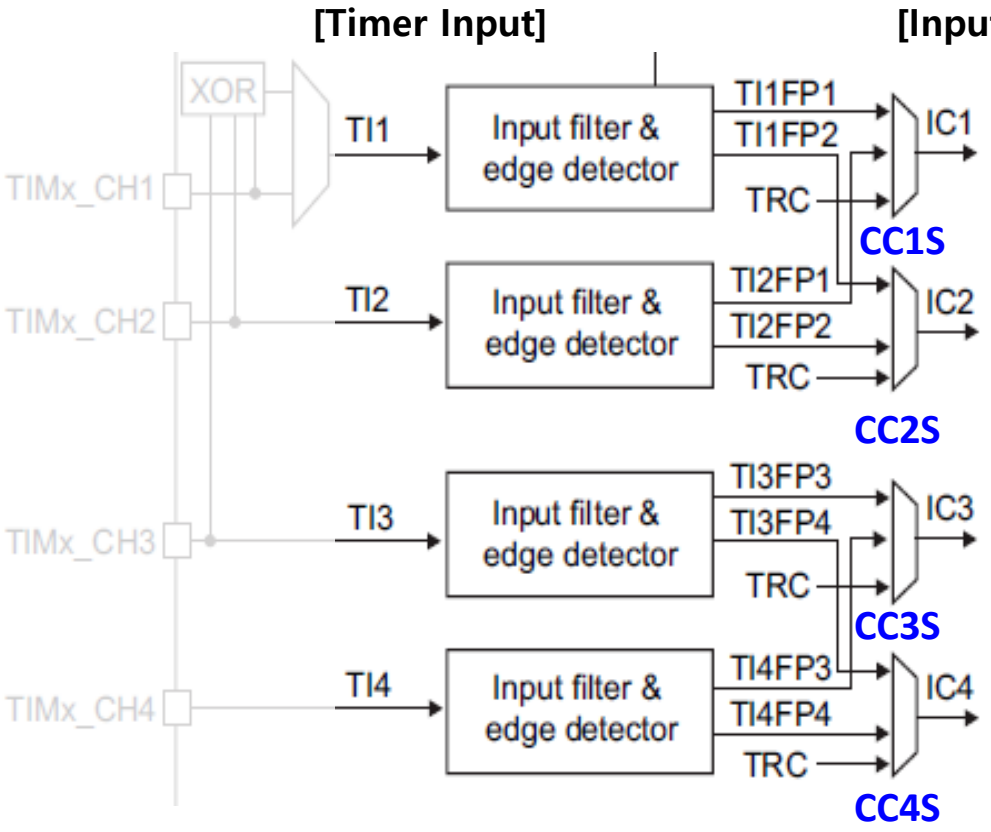
TI4 is from **TIx_CH4**



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Input Capture: Choosing Input

- **Step 1: Choose Time Input Signal**
  - Compare/Capture Selection(**CCyS**): Select TIx for ICy
    - TIMx → CCMRn: CCyS



[TIx – ICy selection]

IC1 can be from **TI1** or **TI2**

IC2 can be from **TI1** or **TI2**

IC3 can be from **TI3** or **TI4**

IC4 can be from **TI3** or **TI4**

TIMx capture/compare mode register 1 (TIMx_CCMR1)

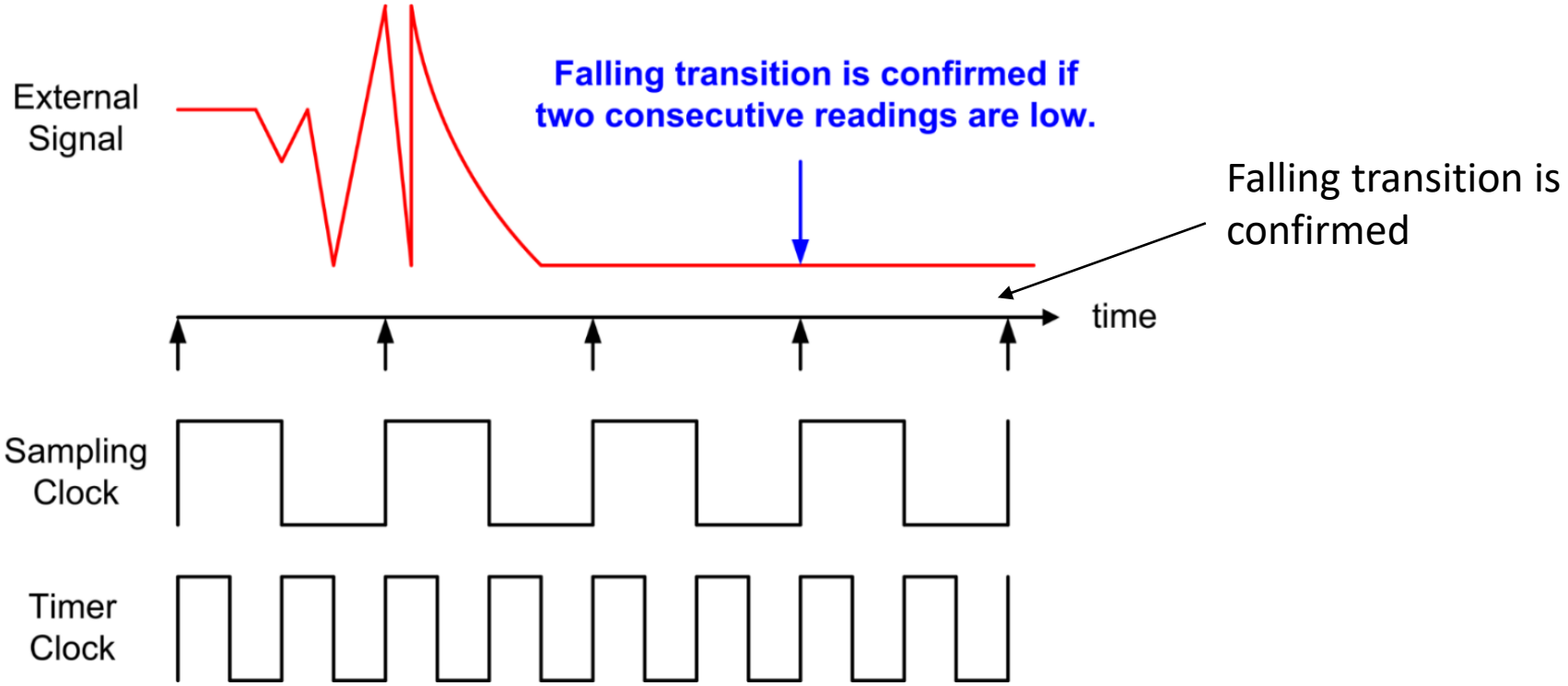
CC1S: Capture/Compare 1 selection

This bit-field defines the direction of the channel (input/output) a
00: CC1 channel is configured as output.
01: CC1 channel is configured as input, IC1 is mapped on TI1.
10: CC1 channel is configured as input, IC1 is mapped on TI2.

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Input Capture

- **Step 2: Choose Input Filter Duration**
  - There can be noises and unstable transient signal
  - One method is wait until input signal is stable
  - Filter Duration: Number of events needed to validate a transition on the input
    - e.g. Falling Edge Detection, Filter Duration N=2



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Input Capture

- **Step 3: Set the Active Edge**
  - Fall / Rise / Fall & Rise

- **Step 4: Set the Prescaler**
  - Frequency Divider. , /1, /2, /4, 8

- **Step 5: Enable Input capture**
  - Enable for each IC channel

- **Step 6: Enable CC+UI Interrupt**
  - Capture Trigger Interrupt (CCIE)
  - Update Interrupt (UIE) : for counting counter overflow (OVC)

  * Two interrupt calls under the same ISR

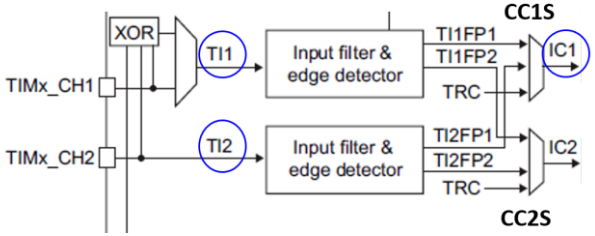
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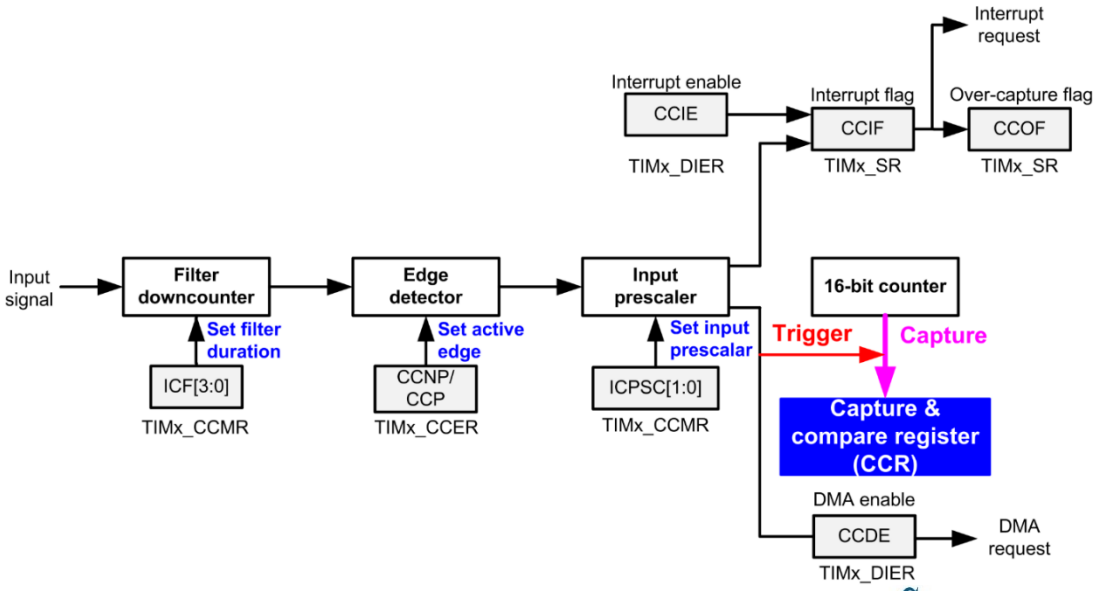
Input Capture: Procedure Summary

Example: Capture with CC_1. TIMx_CH1 for Input Channel_1 at Rising edge signal

1) **Select IC input:** TIMx_CH1 -> TI1 –> CC1S -> IC1
   → TIMx_CCMR1:CC1S=01
2) **Input Capture Filter duration:** Filter N=8
   → TIMx_CCMR1: IC1F=0011(N=8)
3) **Input Prescaler:** PSC=1
   → TIMx_CCMR1: IC1PSC=00
4) **Edge Polarity:** Rising edge
   → TIMx_CCER: CC1NP/CC1P=00
5) **Enable Capture**
   → TIMx_CCER: CC1E=01

TIMx_CCMR1
*TIMx_CCMR2 for CC3~4





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Appendix

Process	Sub-Process	Interrupt (Event)	Interrupt (CCR)	PWM	Input Capture
Clock	System CLK	RCC			
	TIM CLK Enable	RCC_APB1ENR			
Counter Setup	Period	PSC, ARR			
	DIR	DIR			
	Enable	CEN			
Interrupt	Flag	UIF	CCxIF		UIF, CCxOF
	Enable	UIE	CCxE		UIE, CCxE
Compare& Capture	CC value		CCR	CCR	CCR
Output Setup	Pin			GPIO AF	
	Output Mode			OCyM=110 (PWM1) OCyM=111 (PWM2)	
	Polarity			CCxP	
	Enable			CCxE	
Input Setup	Pin				GPIO AF
	IC Select				CCyS
	Filter				ICyF
	Input Pre-scaler				ICyPSC
	Polarity				CCyP
	Enable				CCxE

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Application: Ultrasonic Distance Sensor

- **Ultrasonic distance sensor:**
  - Transmitter: generates bursts of high frequency ultrasonic waves(about 8 pulses of 40KHz)
  - Receiver: detects reflected waves
  - Measure distance from the time difference
    - 2cm (150us) ~ 400cm(25ms)

$$Distance = \frac{Round\ Trip\ Time \times Speed\ of\ Sound}{2}$$

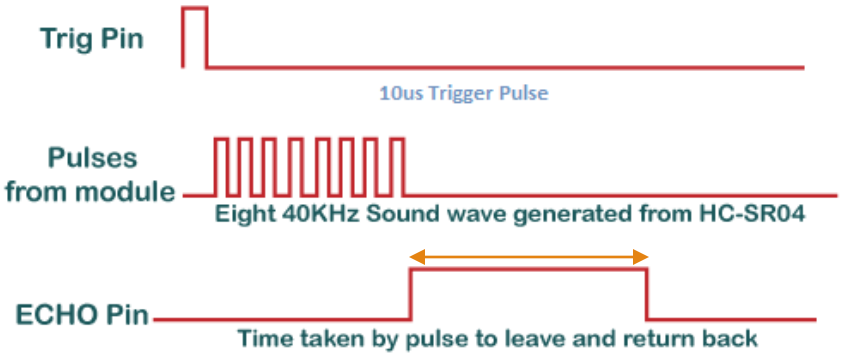
$$= \frac{Round\ Trip\ Time(\mu s) \times 10^{-6} \times 340 m/s}{2}$$

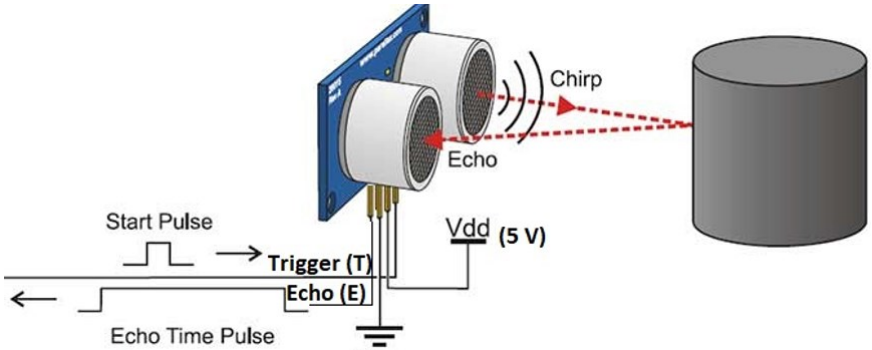
$$= \frac{Round\ Trip\ Time(\mu s)}{58}$$

$$\boxed{Distance\ (cm) = \frac{Pulse\ Width\ (\mu s)}{58}}$$



Ultrasonic HC-SR04 module Timing Diagram





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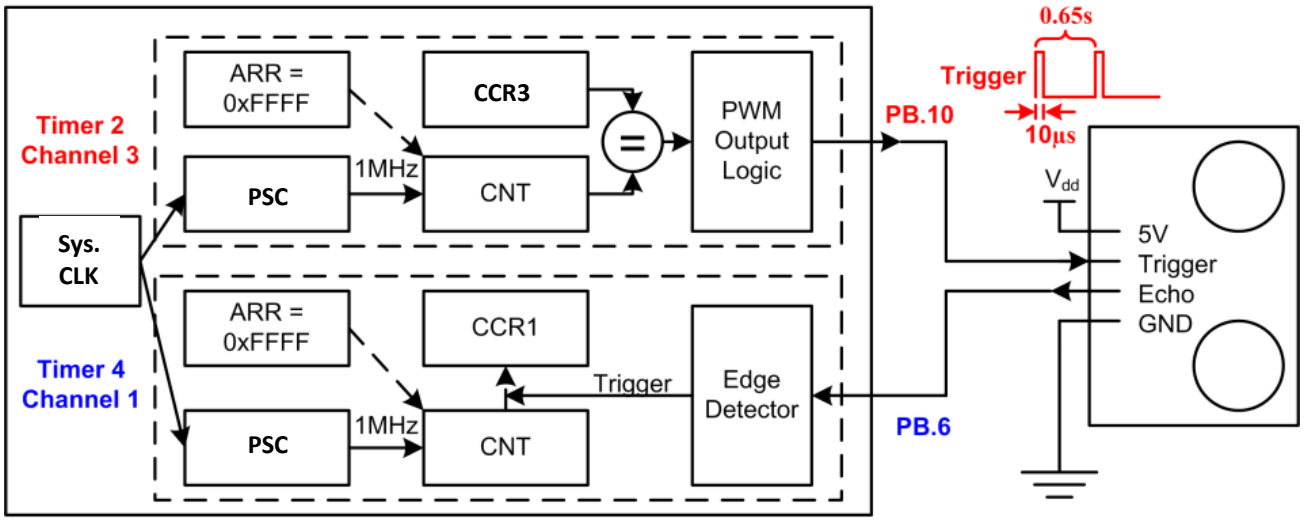
Application: Ultrasonic Distance Sensor

- **Ultrasonic distance sensor** (example)
  - **Trigger Timer (PWM Output)**
    - TIM2_CH3: Trigger signal
    - PWM period : 50 msec
    - PWM duty width: 10us

[Timer 2: 16-bit]
PWM period: 50 msec
→ Calculates CNT_CLK
Pulse width: 10us

  - **Capture Timer (Input Capture)**
    - TIM4_CH1: Input Capture of TI1
    - Counter CLK 1MHz

[Timer 4: 16-bit]
CNT_CLK: 0.1MHz
Overflow period: 10us*2^16=660ms



Edge detector triggers logging the CNT value into CCR1.